PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 09 – Support Vector Machines for Classification and Regression

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Three periods: geometry, kernels, and regression

Connect to Lecture 08

Linear models fitted numbers and class probabilities. SVM instead chooses a score using geometric clearance.

Period	Questions and calculations
1: 45 min	Distance, score normalization, hard-margin calculation, slack, and hinge loss.
2: 45 min	Dual coefficients, KKT, kernels, similarity calculations, and joint parameter selection.
3: 45 min	SVR: tolerance tubes, residual losses, dual prediction, and time-aware regression.

A continuous example

Reuse the same inspection points for geometry and the dual. Then replace class labels by a continuous wear target.

Next lecture removes target labels and asks what a useful group means.

Several separators fit the same training data

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Idea

Each row describes one inspected item: standardized roughness x_1, standardized vibration x_2, and a verified label $y \in \{-1, +1\}$. Correct training signs do not uniquely determine a line.

$$\mathcal{D} = \{((-2, 0), -1), ((-1, 0), -1), ((1, 0), +1), ((2, 0), +1)\}$$

Work through it

The boundaries $x_1 = 0$ and $x_1 = 0.8$ both classify all four rows correctly. Their smallest clearances are 1 and 0.2.

Interpretation

A larger minimum clearance tolerates a larger small measurement shift in the same feature geometry. It is a design principle, not a guaranteed probability of success.

A clearance calculation predicts which perturbations flip a label

Question and idea

Compare the two separators from our inspection example under a measurement change. Keep the fitted boundary fixed: the question is robustness of a prediction, not retraining.

$$x_1 = 1 \to 0.7: \quad f_A = x_1 = 0.7, \quad f_B = x_1 - 0.8 = -0.1$$
$$x_1 = 1 \to 0.9: \quad f_A = 0.9, \quad f_B = 0.1$$

Calculate and explain

A shift of 0.3 flips the second prediction but not the first. A shift of 0.1 leaves both positive. The minimum clearance identifies the first possible sign change along the normal.

Meaning

This is a geometric sensitivity calculation for stated perturbations. It does not supply an error probability or cover arbitrary changes in the data distribution.

A linear score has a direction and a position

Idea

$f(x) = w^T x + b$ is a signed score. The set $f(x) = 0$ is the decision boundary; w is perpendicular to that boundary, while b shifts its location.

$$w = (3,4)^T, \quad b = -5, \quad x = (1,1)^T \quad \Rightarrow \quad f(x) = 3 + 4 - 5 = 2$$

Work through it

A boundary point satisfies $3x_1 + 4x_2 = 5$. Moving parallel to the boundary leaves the score unchanged; moving along w changes it.

Interpretation

A positive score predicts class $+1$. The numerical score 2 is neither a geometric distance nor a probability.

Verify the normal direction with two boundary points

Question and idea

For $3x_1 + 4x_2 = 5$, choose two points on the boundary. Their displacement must be perpendicular to $w = (3, 4)$, whereas movement along w changes the score.

$$u = (1, .5),\ v = (-1, 2), \quad w^T(v - u) = 3(-2) + 4(1.5) = 0$$
$$f(u + tw) = f(u) + t\|w\|^2 = 25t$$

Calculate and explain

Both boundary points have score zero. A step of 0.1 along the unnormalized normal changes the score to 2.5; that step has physical length 0.5 in these coordinates.

Meaning

Score change and travel length are related but are different numbers. A unit normal is needed when the coefficient of a displacement should be a distance.

Derive the distance instead of memorizing it

Idea

Let x_0 be the perpendicular projection of x onto $H = \{z : w^T z + b = 0\}$. Travel along the unit normal $w/\|w\|$.

$$x = x_0 + d\frac{w}{\|w\|} \quad \Rightarrow \quad w^T x + b = d\|w\|$$

$$d = \frac{w^T x + b}{\|w\|}, \qquad \text{dist}(x, H) = |d|$$

Work through it

For $w = (3, 4)$, $b = -5$, and $x = (1, 1)$, the norm is 5: signed distance $= 2/5 = 0.4$.

Interpretation

Keep the sign to identify the side; take the absolute value only when asking for an unsigned distance.

Calculate the projection point and check the residual geometry

Question and idea

The closest boundary point is obtained by subtracting the normal component. For the same w, b, x, use the squared norm in the projection formula.

$$x_0 = x - \frac{w^T x + b}{\|w\|^2} w = (1,1) - \frac{2}{25}(3,4) = (.76, .68)$$

$$3(.76) + 4(.68) - 5 = 0, \qquad \|x - x_0\| = \sqrt{.24^2 + .32^2} = .4$$

Calculate and explain

The projected point lies on the boundary and its distance agrees with the earlier normalized-score calculation. The displacement is parallel to the normal.

Meaning

Subtracting a score times the unit normal would use the wrong scale; the projection coefficient contains the squared norm.

Rescaling a score does not move its boundary

Idea

For any $a > 0$, (w, b) and (aw, ab) give the same boundary and the same signs. A raw functional margin is therefore not an invariant measure of clearance.

$$\frac{y_i((aw)^T x_i + ab)}{\|aw\|} = \frac{a\, y_i(w^T x_i + b)}{a\|w\|} = \frac{y_i(w^T x_i + b)}{\|w\|}$$

Work through it

The previous score 2 becomes 20 after multiplying both parameters by 10, but the distance remains $20/50 = 0.4$.

Interpretation

Compare normalized distances. Never compare two separators solely by the sizes of their raw scores.

Normalize the closest functional margin to one

Idea

For separable data choose the parameter scale so that the smallest labeled score is 1. All training constraints can then use one formula.

$$y_i(w^T x_i + b) \geq 1 \qquad (i = 1, \ldots, N)$$

$$y_i = +1 \Rightarrow f(x_i) \geq 1, \qquad y_i = -1 \Rightarrow f(x_i) \leq -1$$

Work through it

For the four inspection rows, $w = (1, 0)$ and $b = 0$ give labeled scores $2, 1, 1, 2$. The nearest two rows achieve equality.

Interpretation

The constraint value 1 fixes score scale. The physical clearance is still $1/\|w\|$.

Normalize an off-center separator without changing its predictions

Question and idea

The separator $x_1 = 0.8$ classifies all four inspection rows correctly. Its smallest labeled score is 0.2, so multiply both parameters by 5 to meet the canonical constraint.

$$(w, b) = (1, -.8) \to (5, -4)$$

$$x_1 = (-2, -1, 1, 2), \quad y = (-1, -1, 1, 1) \Rightarrow yf = (14, 9, 1, 6)$$

$$\text{half-width} = 1/5 = .2, \qquad \text{full width} = .4$$

Calculate and explain

The boundary remains 0.8 and the signs remain correct. The canonical margin exposes its narrower clearance compared with $w = 1, b = 0$, whose full width is 2.

Meaning

Large normalized scores for distant rows do not compensate for the single closest row that fixes the minimum.

The two supporting planes form a corridor

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Idea

The decision plane is $f(x) = 0$. Its two parallel supporting planes are $f(x) = +1$ and $f(x) = -1$. Under hard-margin constraints, no training point lies inside the open corridor.

$$\text{distance to each supporting plane} = \frac{1}{\|w\|} \qquad \text{full width} = \frac{2}{\|w\|}$$

Work through it

With $w = (1, 0)$ and $b = 0$, the supporting planes are $x_1 = 1$ and $x_1 = -1$: the full corridor is 2 units wide.

Interpretation

Some texts call the half-width the margin. State which convention is used before comparing numerical answers.

Find the midpoint and width from two opposing supports

Question and idea

Suppose the nearest negative and positive observations on one coordinate are 1 and 3. Their supporting equations determine the canonical weight, intercept, and decision location.

$$w(1) + b = -1, \quad w(3) + b = 1 \Rightarrow w = 1,\ b = -2$$
$$f(x) = x - 2, \quad \text{boundary} = 2, \quad \text{full width} = 2$$

Calculate and explain

At $x = 1.6$, the score is-0.4 and the distance is 0.4. The midpoint is appropriate here because the hard-margin optimum is controlled by these symmetric supporting constraints.

Meaning

The midpoint between class averages is a different rule. Hard-margin SVM uses active boundary evidence, not all-class mean locations.

Solve a one-dimensional hard margin by hand

Idea

Keep only the closest opposite-class observations $x_- = -1$ and $x_+ = 1$. Their constraints limit the width that any feasible separator can achieve.

$$-(-w + b) \geq 1, \quad w + b \geq 1 \quad \Rightarrow \quad w \geq 1$$
$$\min \tfrac{1}{2} w^2 \quad \Rightarrow \quad w^* = 1,\ b^* = 0,\ \text{width} = 2$$

Work through it

At the optimum both inequalities are tight. The farther rows -2 and 2 satisfy their constraints without changing this solution.

Interpretation

The optimum balances the nearest opposing evidence. A separator at 0.8 is feasible after scaling, but its margin is smaller.

Solve a shifted pair: separation width changes the optimal norm

Question and idea

Take a negative support at-2 and a positive support at 4. Add their canonical inequalities before minimizing the norm; recover the intercept from the equalities at the optimum.

$$2w - b \geq 1, \quad 4w + b \geq 1 \Rightarrow 6w \geq 2$$

$$w^* = 1/3, \quad b^* = -1/3, \quad \text{boundary} = 1, \quad \text{width} = 6$$

Calculate and explain

A guessed intercept $b = -1$ would give scores-5/3 and 1/3, violating the positive constraint. Substituting the correct intercept-1/3 gives the required scores-1,+1.

Meaning

The optimum is $w = 1/3, b = -1/3$, not the first guessed intercept. Substitution is essential even after the weight is correct.

Hard-margin SVM is a constrained quadratic problem

Idea

After fixing score scale, maximizing corridor width is equivalent to minimizing the squared weight norm. The bias remains a position parameter and is not included in this standard norm penalty.

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1 \quad \forall i$$

Work through it

For D features there are $D+1$ model parameters and N margin constraints. Our hand solution has objective $1/2$ and zero violations.

Interpretation

Convexity gives a global optimum for this problem when feasible; it does not make wrong labels or unrepresentative data reliable.

Compare feasible solutions using both constraints and objective

Question and idea

On the original rows $-2, -1, 1, 2$, both $(w, b) = (1, 0)$ and $(2, 0)$ satisfy hard-margin constraints. The optimization must distinguish these feasible candidates.

$$m_A = (2, 1, 1, 2), \quad m_B = (4, 2, 2, 4)$$
$$J_A = \tfrac{1}{2}(1)^2 = .5, \quad J_B = \tfrac{1}{2}(2)^2 = 2$$

Calculate and explain

Both candidates define the same boundary and have geometric margin 1. The second has larger raw scores and a narrower canonical corridor, but a larger objective. The minimum-norm feasible scaling makes the nearest constraints tight.

Meaning

Raw scores, feasibility, and optimality are three separate statements. Verify feasibility first, then compare the stated norm objective.

Support vectors are the active training constraints

Idea

In a nondegenerate hard-margin solution, critical rows lie on $y_i f(x_i) = 1$. Rows farther away leave room in their inequalities; a small movement of them may leave the optimum unchanged.

$$x = -1, +1 : \ yf(x) = 1; \qquad x = -2, +2 : \ yf(x) = 2$$

Work through it

Move the outer positive row from 2 to 3: our solution remains $w = 1, b = 0$. Move the inner positive row from 1 to 0.6: the optimum must change.

Interpretation

Support does not mean representative, frequent, or certainly correct. It means the row participates in determining the optimized boundary.

Move a critical row and recompute the optimized boundary

Question and idea

Keep the negative support at-1 but move the positive support from 1 to 0.4. This changes the active constraints rather than merely changing a parameter scale.

$$-w + b = -1, \quad .4w + b = 1 \Rightarrow w = 10/7, \quad b = 3/7$$

$$\text{boundary} = -b/w = -.3, \quad \text{width} = 2/w = 1.4$$

Calculate and explain

The separator moves left from 0 to-0.3 and the width shrinks from 2 to 1.4. A distant row moving from 2 to 3 would not cause this change in the same configuration.

Meaning

Support vectors are influential because their constraints are active. Their influence follows from location and optimization, not frequency or representativeness.

Feature units are part of the margin definition

Idea

Euclidean clearance depends on the coordinates supplied to the model. Changing one feature from millimeters to micrometers changes distances unless preprocessing restores a deliberate scale.

$$z_j = \frac{x_j - \mu_j}{s_j}, \qquad \mu_j, s_j \text{ estimated on the training fold only}$$

Work through it

A raw difference $(1, 1000)$ has norm about 1000. If the second coordinate is only a unit conversion of a comparable measurement, it should not automatically dominate similarity.

Interpretation

Scaling is a modeling choice. Save training statistics and reuse them for validation and prediction; do not refit them on held-out rows.

Scaling one coordinate changes the metric, not merely notation

Question and idea

Let a coordinate transformation be $z = Sx$. Euclidean distance in z corresponds to a weighted distance in x; anisotropic scaling therefore changes which directions count as large perturbations.

$$S = \mathrm{diag}(1, 10), \quad \Delta x = (1, 1)$$
$$\|\Delta x\| = \sqrt{2}, \qquad \|S\Delta x\| = \sqrt{101} \approx 10.05$$

Calculate and explain

A tenfold unit expansion in the second coordinate increases the contribution of that direction. A scaler or meaningful feature weights must make the intended comparison explicit.

Meaning

A rotation preserves Euclidean norm; unequal coordinate scaling does not. Normalizing the classifier score cannot undo an inappropriate input metric.

Perfect separation can follow one unreliable row

Idea

A mislabeled or unusual observation can make hard-margin constraints infeasible, or force a very narrow corridor. Fitting all training signs is then in conflict with a stable boundary.

$$x = 1,\ y = +1 \quad \text{and} \quad x = 1,\ y = -1 \quad \Rightarrow \quad f(1) \geq 1,\ f(1) \leq -1$$

Work through it

The two conflicting rows cannot both satisfy hard-margin constraints. A penalty for violations lets the model choose a compromise instead of demanding an impossible fit.

Interpretation

Soft margin does not repair a known labeling error. Correct the data when possible; otherwise model the uncertainty explicitly.

Conflicting duplicate labels reveal the role of costs

Question and idea

At one identical input location, a positive and negative label receive the same score f. Their combined hinge loss has an irreducible floor.

$$\max(0, 1 - f) + \max(0, 1 + f) = 2 \quad \text{for } -1 \le f \le 1$$

$$2\max(0, 1 - f) + \max(0, 1 + f) \text{ is minimized at } f = 1$$

Calculate and explain

Equal row costs cannot distinguish scores inside the interval by data loss alone. Doubling the positive cost prefers its margin, leaving a negative-row loss of 2.

Meaning

Costs choose how to resolve unavoidable conflict. They do not create new information that separates identical inputs with opposite labels.

Slack measures a margin violation, not just an error

Idea

Write $m_i = y_i f(x_i)$. The smallest feasible slack is $\xi_i = \max(0, 1 - m_i)$, so being inside the corridor and being misclassified are different events.

m_i	ξ_i	Sign correct?	Position
1.4	0	yes	outside/on margin
0.4	0.6	yes	inside corridor
0	1	boundary	decision plane
-0.3	1.3	no	wrong side

Work through it

The row with slack 0.6 is penalized even though its predicted class is correct. The row with slack 1.3 crosses the decision boundary.

Interpretation

Slack is measured in normalized score units. It is not automatically a Euclidean distance.

Total slack upper-bounds errors but is not their count

Question and idea

Calculate the minimal slacks for four labeled scores and compare them with the number of incorrect signs. A correctly classified point can still pay a margin penalty.

$$m = (2, 1, .5, -1) \Rightarrow \xi = (0, 0, .5, 2), \quad \sum_i \xi_i = 2.5$$

incorrect signs $= 1$ because only $m = -1$ is negative

Calculate and explain

The third row is correct but contributes 0.5 slack; the last row contributes 2 for one classification error. Total violation records severity as well as sign mistakes.

Meaning

An error has slack at least 1 under this convention, so total slack bounds its count. It is not a calibrated error rate or a one-to-one error tally.

Compare two compromises with the same objective

Idea

Soft-margin SVM balances a short weight vector against total violation. C is the price assigned to one unit of slack in the chosen objective convention.

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_i \xi_i \quad \text{s.t. } y_i f(x_i) \geq 1 - \xi_i,\ \xi_i \geq 0$$

$$J_A = 0.5 + 1.2C, \quad J_B = 2 + 0.2C$$

Work through it

For two hypothetical feasible candidates, $C = 0.5$ gives $J_A = 1.1 < J_B = 2.1$; $C = 3$ gives $J_B = 2.6 < J_A = 4.1$.

Interpretation

Increasing C changes the preferred tradeoff. It does not promise a monotonic improvement in validation accuracy.

Doubling identical rows doubles the summed-loss price

Question and idea

If every training row is duplicated, a summed-loss objective doubles the fit term while keeping the norm term fixed. Identical independent evidence has not actually doubled.

$$J = \tfrac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \ell_i \quad \rightarrow \quad \tfrac{1}{2}\|w\|^2 + 2C\sum_{i=1}^{N} \ell_i$$

Calculate and explain

To preserve the original tradeoff after exact duplication, use $C_{\text{new}} = C/2$. Under an averaged-loss convention, duplicating every row leaves the fit average unchanged.

Meaning

Unequal duplication acts as row weighting. Record both sample counts and loss convention before comparing parameter values between experiments.

Eliminate slack and recover hinge loss

Idea

For fixed w, b, positive violation costs make the optimal slack as small as possible. Substituting it gives a loss-plus-regularization objective.

$$\ell_{\text{hinge}}(y, f) = \max(0, 1 - yf)$$

$$J = \frac{1}{2}\|w\|^2 + C \sum_i \max(0, 1 - y_i f(x_i))$$

Work through it

Scores $m = (1.4, 0.4, -0.3)$ produce losses $(0, 0.6, 1.3)$ and total hinge loss 1.9. Zero-one loss would count only the last row as an error.

Interpretation

Hinge loss rewards correct signs with sufficient margin. Logistic loss keeps decreasing smoothly even beyond that margin.

Take one hinge-loss subgradient step and inspect the score

Question and idea

For one positive row $x = (2, 1)$, start with $w = 0, b = 0$, $C = 1$. Its margin is below 1, so the hinge contribution has gradients $-yx$ and $-y$.

$$\eta = .1: \quad w^{\text{new}} = .1(2, 1) = (.2, .1), \quad b^{\text{new}} = .1$$
$$f^{\text{new}}(x) = .2(2) + .1(1) + .1 = .6, \quad \ell^{\text{new}} = .4$$

Calculate and explain

The initial hinge loss was 1. The step reduces this row's loss to 0.4 but does not yet achieve the unit margin. The norm gradient was zero only at the starting point.

Meaning

Later updates must include regularization and all relevant rows. Beyond the margin, this row has zero hinge gradient rather than an endless demand for larger score.

Weights and parameter conventions change the tradeoff

Idea

If missing a defect is more costly, use row or class weights in the violation term. Distinguish this cost choice from the later choice of a prediction threshold.

$$J = \tfrac{1}{2}\|w\|^2 + C\sum_i a_i \ell_i \qquad (a_i > 0)$$

$$\tfrac{\lambda}{2}\|w\|^2 + \tfrac{1}{N}\sum_i \ell_i \quad \Longleftrightarrow \quad C = \frac{1}{N\lambda}$$

Work through it

With $N = 100$ and $\lambda = 0.02$, the equivalent unweighted summed-loss convention has $C = 0.5$. Duplicating rows also increases their effective influence.

Interpretation

A bigger C means a smaller equivalent regularization strength here. Compare conventions before copying a number from another model.

Balanced class weights equalize total class influence

Question and idea

For 100 rows with 3 defects and 97 acceptable items, inverse-frequency class weights can give each class the same total loss weight. They do not equalize individual-row importance.

$$a_+ = \frac{100}{2 \cdot 3} \approx 16.67, \quad a_- = \frac{100}{2 \cdot 97} \approx .515$$
$$3a_+ = 50, \qquad 97a_- = 50$$

Calculate and explain

One defect row receives about 32.3 times an acceptable row's multiplier. This compensates frequency under this chosen policy, while a separate economic cost may require different weights.

Meaning

Class balancing is not automatically the application-optimal cost. Evaluate recall, false alarms, and the intended decision threshold together.

Why rewrite the model using training rows?

Idea

The primal searches over feature coefficients w. The dual searches over row coefficients α_i and expresses the solution through inner products between observations.

$$w = \sum_i \alpha_i y_i x_i \quad \Rightarrow \quad f(x) = \sum_i \alpha_i y_i \, x_i^T x + b$$

Work through it

A feature vector may have thousands of coordinates, yet prediction can be described by similarities to the support rows. For a linear model the sum can be compressed back into one w.

Interpretation

The dual is not always computationally cheaper. Its main teaching role here is explaining sparsity and opening a route to kernels.

Distinguish feature sparsity from support-vector sparsity

Question and idea

A coefficient vector w has one entry per feature; dual coefficients have one entry per training row. Sparse features and a small support set are different ways to simplify a prediction.

$$w = (0, 2, 0, -1): \text{ 2 nonzero feature coefficients}$$
$$\alpha = (0, .4, 0, 0, .7, 0): \text{ 2 active training rows}$$

Calculate and explain

The first statement selects coordinates; the second selects sample contributions. A kernel SVM can have many active supports even when every input vector is sparse.

Meaning

The usual L 2 margin penalty does not automatically make w sparse. Nonzero alpha identifies stored row evidence, not selected input variables.

Put both soft-margin constraints into a Lagrangian

Idea

Write constraints as $1 - \xi_i - y_i f(x_i) \leq 0$ and $-\xi_i \leq 0$. Attach nonnegative multipliers α_i and μ_i respectively.

$$L = \frac{1}{2}\|w\|^2 + C\sum_i \xi_i + \sum_i \alpha_i(1 - \xi_i - y_i(w^T x_i + b)) - \sum_i \mu_i \xi_i$$

Work through it

If a margin constraint is tight, its multiplier may be positive. A strictly inactive constraint must have zero multiplier at the optimum.

Interpretation

Keep every sign consistent with the chosen inequality direction. We need only the SVM-specific stationarity and complementarity conditions.

A violated constraint has a positive multiplier term

Question and idea

With constraints written as $g_i = 1 - \xi_i - m_i \leq 0$, check a numerical row before reading the full Lagrangian. A positive g_i means the row is still infeasible.

$$m_i = .4,\ \xi_i = .3: \quad g_i = 1 - .3 - .4 = .3 > 0$$
$$\alpha_i = .2: \quad \alpha_i g_i = .06; \qquad \xi_i = .6 \Rightarrow g_i = 0$$

Calculate and explain

The smaller slack fails the margin constraint; increasing it to 0.6 makes the constraint tight. A nonnegative multiplier charges rather than rewards positive violation in this inequality convention.

Meaning

Changing the inequality direction requires changing the sign convention consistently. Constraint feasibility is checked independently of whether a score has the correct sign.

Three derivatives reveal three pieces of structure

Idea

Minimize the Lagrangian over primal variables. Stationarity eliminates w, b, ξ and leaves a constrained optimization in the row multipliers.

$$\nabla_w L = 0 \Rightarrow w = \sum_i \alpha_i y_i x_i$$

$$\partial_b L = 0 \Rightarrow \sum_i \alpha_i y_i = 0$$

$$\partial_{\xi_i} L = 0 \Rightarrow C - \alpha_i - \mu_i = 0 \Rightarrow 0 \le \alpha_i \le C$$

Work through it

The first line reconstructs a direction, the second balances the signed coefficients, and the third limits the influence of any one row under this loss price.

Interpretation

The upper bound comes from soft-margin penalties. Hard-margin multipliers have no finite C cap.

Row-dependent costs create row-dependent coefficient caps

Question and idea

Replace C by a row price $C_i = Ca_i$. The slack stationarity equation gives a separate upper bound for each row, directly connecting class weights with dual influence.

$$\partial_{\xi_i} L = C_i - \alpha_i - \mu_i = 0 \Rightarrow 0 \le \alpha_i \le C_i$$

$$C = 2,\ a_+ = 3,\ a_- = 1: \quad \alpha_+ \le 6, \quad \alpha_- \le 2$$

Calculate and explain

A positive row can reach a cap of 6 while a negative row reaches 2. Label balance still requires the signed sum of all alpha values to vanish.

Meaning

A larger cap permits greater influence but does not force its coefficient to reach that cap. KKT determines whether the row is active.

Read the dual objective by its two terms

Idea

Substitution produces a concave quadratic maximization. A linear reward increases row coefficients; a quadratic term charges for the resulting boundary direction.

$$\max_{\alpha} \sum_{i} \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$\text{s.t. } 0 \leq \alpha_i \leq C, \qquad \sum_{i} \alpha_i y_i = 0$$

Work through it

The data enter only through $x_i^T x_j$. The $N \times N$ Gram matrix stores these inner products; labels determine their signed contribution.

Interpretation

The primal and dual describe the same convex SVM optimum under the usual feasibility conditions. Rewriting does not create extra labeled data.

Expand the two-point Gram matrix and labeled quadratic form

Question and idea

For supports $x_- = -1$ and $x_+ = 1$, distinguish the input Gram matrix from the label-adjusted matrix in the dual. Opposite input signs and opposite labels cancel.

$$K = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad Y = \mathrm{diag}(-1, 1), \quad YKY = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\alpha = (.5, .5): \quad \alpha^T YKY\alpha = 1, \quad W = 1 - .5 = .5$$

Calculate and explain

The dual reward is the alpha sum 1; the quadratic charge is half of 1. This matches the squared norm of the reconstructed weight $w = 1$.

Meaning

A negative raw inner product is not itself invalid. The entire Gram matrix is the relevant positive-semidefinite object.

KKT separates inactive rows, free supports, and violations

Idea

Complementarity is $\alpha_i(1-\xi_i-m_i)=0$ and $(C-\alpha_i)\xi_i=0$, with $m_i=y_i f(x_i)$. These conditions explain which rows can influence the optimized model.

Coefficient	Typical margin state	Meaning
$\alpha_i=0$	$m_i \geq 1$	no prediction term
$0<\alpha_i<C$	$m_i=1, \xi_i=0$	free support vector
$\alpha_i=C$	$m_i \leq 1$	on/inside margin, possibly wrong

Work through it

An at-bound coefficient is not necessarily a misclassification: $m_i=0.4$ is correct but violates the margin.

Interpretation

Boundary equalities can occur in degenerate cases. Use actual nonzero coefficients to identify supports in software.

Check a complete KKT row state rather than one inequality

Question and idea

Use $C = 1$ and $m_i = y_i f(x_i)$. Each candidate state must satisfy nonnegativity, the margin constraint, slack stationarity, and both complementarity products.

m_i	ξ_i	α_i	$\mu_i = C - \alpha_i$
1.5	0	0	1
1	0	.4	.6
-.2	1.2	1	0

Calculate and explain

For the middle row, the margin residual and slack are both zero. For the last row, the margin residual is zero and the nonnegativity multiplier is zero, allowing positive slack.

Meaning

Inspecting alpha alone is insufficient to infer whether a row is wrong. Its labeled score and slack determine that interpretation.

Recover the intercept from a free support vector

Idea

For $0 < \alpha_i < C$, KKT gives $y_i f(x_i) = 1$. Since $y_i^2 = 1$, this row supplies an intercept estimate; average across free supports for numerical stability.

$$b_i = y_i - \sum_j \alpha_j y_j x_j^T x_i \qquad (0 < \alpha_i < C)$$

Work through it

In our two-support solution, $w = 1$: the positive row gives $b = 1 - 1 = 0$, and the negative row gives $b = -1 - (-1) = 0$.

Interpretation

Do not apply this equality to an arbitrary at-bound support with positive slack. If there are no free supports, use feasible KKT bounds for b.

Solve the same two-point problem in the dual

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Idea

Take $x_- = -1, y_- = -1$ and $x_+ = 1, y_+ = 1$, with $C > 1/2$. Label balance requires $\alpha_- = \alpha_+ = a$.

$$w = a(-1)(-1) + a(+1)(+1) = 2a$$
$$W(a) = 2a - \tfrac{1}{2}(2a)^2 = 2a - 2a^2$$
$$W'(a) = 2 - 4a = 0 \Rightarrow a = \tfrac{1}{2}, \quad w = 1, \quad b = 0$$

Work through it

The dual optimum is $W = 1/2$, matching the primal objective $\|w\|^2/2 = 1/2$. Both samples carry coefficient $1/2$.

Interpretation

This calculation verifies the representation. If $C < 1/2$, the coefficient cap changes the solution and slack is needed.

At small C, the same two supports move inside the margin

Question and idea

In the two-point dual $W(a) = 2a - 2a^2$, label balance gives equal coefficients. If $C = 0.2$, the unconstrained optimum 0.5 is beyond the cap.

$$a^* = .2, \quad w = 2a = .4, \quad b = 0, \quad m_- = m_+ = .4$$
$$\xi_- + \xi_+ = 1.2, \quad J = .5(.4)^2 + .2(1.2) = .32 = W(.2)$$

Calculate and explain

Both signs are correct, yet both rows violate the unit margin and have at-bound coefficients. The formal corridor width is 5, with training rows inside it.

Meaning

Soft-margin width is not a guaranteed empty gap. Lower norm can deliberately accept training-margin violations when their price is small.

Prediction is a signed similarity sum

Idea

Only rows with nonzero α_i contribute. Their labels set the sign of each contribution, while the coefficient sets its strength.

$$f(x) = \sum_{i \in \mathrm{SV}} \alpha_i y_i x_i^T x + b$$

$$f(0.6) = \tfrac{1}{2}(-1)(-1 \cdot 0.6) + \tfrac{1}{2}(+1)(1 \cdot 0.6) = 0.6$$

Work through it

For $x = -0.2$, the score is -0.2, so prediction is class -1. The same linear sum could be stored as $w = 1, b = 0$.

Interpretation

Signed contributions are not individual probabilities or votes. Kernelization changes the similarity calculation, not the need to sum them.

SMO updates a pair to preserve label balance

Idea

Changing one nonzero coefficient alone usually violates $\sum_i \alpha_i y_i = 0$. Sequential minimal optimization fixes all other coefficients and optimizes a feasible pair.

$$y_i \Delta\alpha_i + y_j \Delta\alpha_j = 0$$
$$y_i = -1,\ y_j = +1 \Rightarrow \Delta\alpha_i = \Delta\alpha_j$$

Work through it

With opposite labels and $C = 1$, a pair starting at $(0.2, 0.2)$ may move to $(0.4, 0.4)$ while preserving balance; moving only the second to 0.4 breaks it.

Interpretation

The pair optimum is clipped to its feasible interval. Working-set selection uses KKT violations; numerical tolerance controls when updates stop.

Same-label and opposite-label pairs have different feasible moves

Question and idea

The balance constraint requires $y_i \Delta\alpha_i + y_j \Delta\alpha_j = 0$. With equal labels, their coefficient sum is fixed; with opposite labels, their coefficient difference is fixed.

$$y_i = y_j = 1: \quad \alpha_i + \alpha_j = .8$$

$$C = 1: \quad (.2, .6) \to (.7, .1) \text{ is feasible; } (.3, .7) \text{ is not}$$

Calculate and explain

The first move preserves the sum 0.8 and stays in both bounds. The second raises the sum to 1.0, violating balance even though each coefficient is individually legal.

Meaning

Pairwise optimization respects a line segment inside the coefficient box. Checking only $0 \leq \alpha \leq C$ misses the equality constraint.

Choose a solver from the size of rows and features

Representation and scale

The primal has feature parameters; the dual has row coefficients. Sample count N, feature count D, sparsity, and implementation determine the useful route.

Setting	Starting point
Huge N, moderate D	A scalable linear primal, coordinate, or stochastic solver.
Sparse text	Linear SVM with sparse matrices; preserve sparse input.
Moderate N, nonlinear structure	Kernel SVC after checking memory and validation benefits.

Estimate before fitting

A full double-precision Gram matrix at $N = 20{,}000$ needs $N^2 \times 8 \approx 3.2$ GB, before other storage. Doubling N quadruples this matrix.

Solver choice changes computational work, not the task or the required validation evidence.

A feature map can turn rings into a threshold

Read the figure

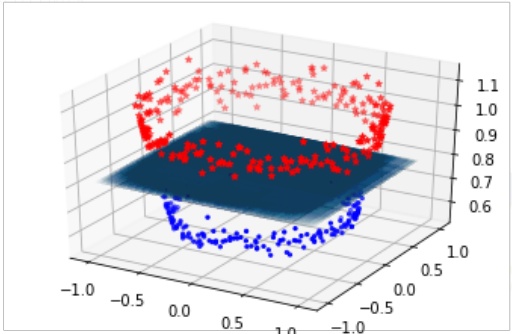
The reference illustrates a lifted ring pattern. For a separate radial hand example, add $z = x_1^2 + x_2^2$ and choose the score $f(x) = z - 1$. A threshold in z is a circle in the original plane.

Follow the evidence

Our hand example gives $f(0.5, 0) = 0.25 - 1 = -0.75$ and $f(1.5, 0) = 2.25 - 1 = 1.25$. The score is linear in the lifted coordinate, although curved in the original inputs.

Interpretation

A feature map encodes a shape hypothesis. The reference plot motivates the mechanism; its plotted coordinates are not this separate numerical example.



The kernel trick computes a mapped inner product

Idea

A kernel $K(x, z) = \phi(x)^T \phi(z)$ supplies exactly the quantity required by the dual and prediction. We may compute it without explicitly forming every mapped coordinate.

$$f(x) = \sum_{i \in \mathrm{SV}} \alpha_i y_i K(x_i, x) + b$$

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

Work through it

With the linear kernel $K(x, z) = x^T z$, these equations reduce to the model we already solved. A nonlinear kernel changes the geometry of similarity.

Interpretation

The trick avoids explicit features, not training data dependence. Prediction still needs kernels to retained support vectors.

Expand a quadratic kernel and verify one calculation

Idea

For two-dimensional inputs, $K(x, z) = (1 + x^T z)^2$ includes squares and an interaction. The square-root factors preserve the exact inner product.

$$\phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1x_2, x_2^2)$$
$$x = (1, 2),\ z = (2, 1): \quad K = (1 + 4)^2 = 25$$
$$\phi(x)^T \phi(z) = 1 + 4 + 4 + 4 + 8 + 4 = 25$$

Work through it

The product x_1x_2 lets a linear classifier in the mapped space express joint feature effects. The kernel computes the same result with one short formula.

Interpretation

Kernel degree is selected using validation. Higher degree increases expressive choices and can amplify scale differences.

Construct a composite kernel by concatenating feature maps

Question and idea

Add the linear kernel to the inhomogeneous quadratic kernel. Concatenating their feature maps produces exactly the sum of their inner products.

$$K(x,z) = (1 + x^T z)^2 + x^T z$$

$$x = 1, z = 2: \quad K = (1+2)^2 + 2 = 11$$

$$\psi(x) = (1, \sqrt{2}x, x^2, x), \quad \psi(1)^T \psi(2) = 1 + 4 + 4 + 2 = 11$$

Calculate and explain

The extra linear term duplicates a direction with a different effective weight; it does not create a separate data source. Nonnegative weighted sums preserve validity.

Meaning

Combining kernels still requires validation of the chosen similarity geometry. A valid mathematical kernel need not be useful for the task.

A valid kernel must have positive-semidefinite Gram matrices

Idea

An inner-product kernel is symmetric. For every finite sample set its Gram matrix must satisfy $v^T K v \geq 0$ for every coefficient vector v.

$$K = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad v = (1, -1)^T \quad \Rightarrow \quad v^T K v = -2 < 0$$

Work through it

This symmetric matrix fails the inner-product test. In contrast, nonnegative sums and products of valid kernels remain valid; explicit feature maps also construct valid kernels.

Interpretation

Symmetry alone is insufficient. A visually plausible similarity or an arbitrary sigmoid parameter choice may fail validity.

RBF similarity has a distance scale you can calculate

Idea

The radial basis kernel compares squared distance after the chosen preprocessing. γ sets how quickly similarity falls as two rows separate.

$$K(x, z) = \exp(-\gamma \|x - z\|^2)$$

$$\gamma = 0.5: \quad d = 0 \Rightarrow 1, \quad d = 1 \Rightarrow 0.607, \quad d = 2 \Rightarrow 0.135$$

Work through it

For inspection rows $(0, 0)$ and $(1, 1)$, squared distance is 2, so the kernel is $e^{-1} \approx 0.368$. Doubling both coordinate differences gives $e^{-4} \approx 0.018$.

Interpretation

A kernel value is a similarity, not the posterior probability that two rows have the same label.

An anisotropic RBF assigns different widths to different coordinates

Question and idea

Instead of one shared gamma, choose nonnegative coordinate weights. This is a radial kernel after a corresponding coordinate rescaling and defines an elliptical neighborhood in the raw coordinates.

$$K(x, z) = \exp\left(-\sum_j \gamma_j (x_j - z_j)^2\right)$$

$$\Delta x = (1, 3), \quad \gamma = (.5, .02): \quad K = e^{-(.5+.18)} = e^{-.68} \approx .507$$

Calculate and explain

A shared gamma 0.5 would give $e^{-5} \approx 0.0067$ for the same pair. The second coordinate is treated as much more tolerant in the anisotropic model.

Meaning

The width choice encodes a prior about relevant variation. Tune it inside the same preprocessing and validation process; more parameters can overfit.

Gamma controls locality, while bandwidth controls width

Idea

Write the same RBF kernel as $\exp(-\|x - z\|^2/(2\sigma^2))$. Then $\gamma = 1/(2\sigma^2)$: larger gamma means a narrower neighborhood.

$$K = e^{-1} \text{ at } d = \frac{1}{\sqrt{\gamma}} \qquad \gamma = 0.25 \Rightarrow d = 2; \quad \gamma = 4 \Rightarrow d = 0.5$$

Work through it

At distance 1, gamma 0.25 gives similarity 0.779, while gamma 4 gives 0.018. The same pair can be strongly or weakly connected under different widths.

Interpretation

Locality changes boundary flexibility. A narrower kernel can fit small islands, but noise may create those islands too.

C and gamma control different parts of flexibility

Idea

Gamma determines the locality of similarities; C determines the price of training margin violations. Tune them jointly because one may partly compensate for the other.

	Small C	Large C
Small γ	wide influence, tolerant fit	wide influence, costly violations
Large γ	local influence, tolerant fit	local influence, costly violations

Work through it

A high-C, high-gamma model can place narrow islands around rows and fit nearly every training label. A small-gamma model may need a larger C to express a useful boundary.

Interpretation

These are structural tendencies. The selected combination must be supported by validation under the intended sampling process.

Read a validation heatmap as a region of evidence

Read the figure

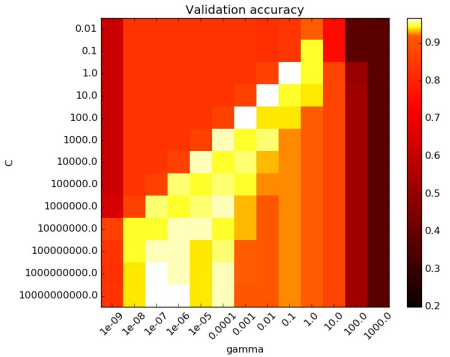
Each cell corresponds to one (C, γ) pair evaluated with the same folds and preprocessing. Read axis direction and color scale before locating a high-scoring region.

Follow the evidence

Compare neighboring cells, not only the brightest one. A broad useful region suggests less sensitivity to the exact parameter choice; a sharp isolated peak deserves closer inspection.

Interpretation

The reference heatmap illustrates interaction on its dataset. Its best cell is not a universal default for inspection data.



Put scaling inside each validation fold

Idea

Search over a pipeline, not over a classifier trained on globally transformed data. Every training fold fits its own scaler and SVM; the paired validation fold receives only their transforms and predictions.

$$\text{training fold} \xrightarrow{\text{fit scaler}} z \xrightarrow{\text{fit SVM}} f$$

$$\text{validation fold} \xrightarrow{\text{saved scaler}} z_{\text{val}} \xrightarrow{f} \text{score}$$

Work through it

Use grouped folds if rows share an inspected item or machine batch. Select parameters from validation, refit on development data, and evaluate the held-out test once.

Interpretation

A visually clean margin cannot compensate for a validation split that allows near-duplicate items on both sides.

Fit preprocessing on each fold: a numerical leakage contrast

Question and idea

Assume a training fold supplies mean 2 and scale 1. A held-out reading 10 must use those values. A transform refitted using later data answers a different experiment.

$$z_{\text{heldout}} = (10 - 2)/1 = 8$$

$$\text{hypothetical globally fitted } \mu = 6, s = 4: \quad z = (10 - 6)/4 = 1$$

Calculate and explain

The first transform sees an extreme future reading; the second has already adapted the coordinate system to later observations. These stated statistics illustrate why a fixed classifier receives different input geometry.

Meaning

The numerical contrast is not an estimate from a claimed dataset. The test of performance requires holding the whole fitted transform fixed, not merely holding SVM coefficients fixed.

Multiclass SVM combines binary decision problems

Idea

For M categories, one-vs-rest trains M binary models; one-vs-one trains $M(M-1)/2$ pairwise models. Their combination rules are part of the estimator.

$$M = 4: \qquad \text{OvR} = 4 \text{ models}, \qquad \text{OvO} = 6 \text{ models}$$

Work through it

For three defect types, OvR compares type A with all others, and similarly for B and C. OvO instead compares A/B, A/C, and B/C using the corresponding subsets.

Interpretation

Raw decision scores need a stated combination rule. They are not a vector of probabilities merely because one score is returned per class.

Calibrate probabilities only when the task needs them

Idea

SVM optimizes a margin loss rather than class likelihood. A second model can map scores to empirical probabilities using separate or out-of-fold calibration data.

$$\widehat{p}(y = +1 \mid f) = \frac{1}{1 + \exp(Af + B)}$$

Work through it

If two systems output scores 2 and 20 for the same geometric boundary after rescaling, neither score alone is an error probability. Calibration learns how scores relate to observed outcomes.

Interpretation

Ranking, classification, and probability estimation need different checks. Do not use held-out test labels to fit a calibration map.

Translate the experiment into a short, explicit implementation

One experiment, one fitted pipeline

Search the scaler and SVM together. The grouped splitter keeps rows from the same machine batch on one side of each fold.

```
from sklearn import svm, preprocessing, pipeline, model_selection
model = pipeline.make_pipeline(preprocessing.StandardScaler(), svm.SVC())
grid = {"svc__C": [0.1, 1, 10], "svc__gamma": [0.01, 0.1, 1]}
search = model_selection.GridSearchCV(
    model, grid, cv=group_folds, scoring="balanced_accuracy")
search.fit(X_dev, y_dev, groups=batch_id)
```

Interpret the choices

Nine candidate pairs share the same folds and metric. The selected object retains its trained scaler; test rows receive only its saved transform and predictions.

These grid values are starting candidates. Sparse text needs a compatible scaling choice; untouched test data evaluate the final selection.

Support-vector count translates into prediction work

Idea

A kernel prediction evaluates n_{SV} similarities and combines them. For dense D-dimensional rows, a direct RBF calculation takes work proportional to $n_{\mathrm{SV}} D$.

$$n_{\mathrm{SV}} = 500,\ D = 40: \quad 500 \times 40 = 20{,}000 \text{ coordinate comparisons per query}$$

Work through it

A linear SVM can precompute w and use only D products per query. Kernel approximations can replace a very large support expansion when scale becomes the bottleneck.

Interpretation

Count, latency, and validation error answer different questions. A small support set does not prove better generalization.

Regression replaces the corridor with an error tube

Read the figure

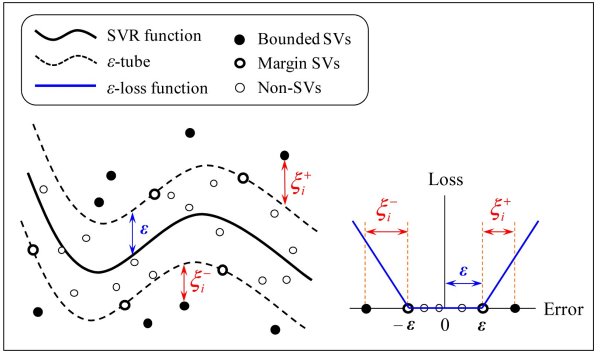
Support vector regression predicts a continuous target. Around $f(x)$ place a tube of half-width ϵ: residuals within the tube contribute no data-fit loss.

Follow the evidence

If an inspection instrument tolerates a prediction error of 0.2 output units, use that as an interpretable candidate epsilon. A residual 0.15 is inside; 0.5 exceeds the tube by 0.3.

Interpretation

The tube is measured vertically in target units, not by perpendicular distance in the input space. The full vertical width is 2ϵ.



Give epsilon an output-unit meaning before tuning it

Question and idea

Epsilon is measured in target units. For wear measured in millimeters, a tolerance 0.1 mm and 1 mm represent very different requirements, even if both are familiar decimal parameter values.

$$y = 2.0 \text{ mm}, \quad f = 2.6 \text{ mm}, \quad |r| = .6 \text{ mm}$$
$$\ell_{.1} = .5 \text{ mm}, \qquad \ell_1 = 0$$

Calculate and explain

The same prediction is outside the narrow tube and inside the wide one. Measurement repeatability, acceptable decision error, and fitted residuals may inform candidate tolerances.

Meaning

Instrument noise and operational tolerance are related considerations, not automatically the same quantity. Report the selected tube in original output units.

Calculate epsilon-insensitive loss on real residuals

Idea

Let $r = y - f(x)$. The loss is zero inside the tube and grows linearly with only the excess beyond its edge.

$$\ell_\epsilon(r) = \max(0, |r| - \epsilon)$$

$$\epsilon = 0.2: \quad r = (0.1, -0.2, 0.5, -0.7) \Rightarrow \ell = (0, 0, 0.3, 0.5)$$

Work through it

The total epsilon loss is 0.8. Absolute loss on the same residuals is 1.5; squared loss is 0.79. These values optimize different preferences and have different units.

Interpretation

Epsilon ignores small residuals; it does not remove the regularization penalty or declare measurement errors nonexistent.

A constant predictor can have zero tube loss without exact fit

Question and idea

Three observations have targets 1.0,1.1,1.2. Choose $f(x) = 1.1$ and epsilon 0.1. Their residuals are nonzero for two rows but remain within the tube.

$$r = (-.1, 0, .1), \quad \ell_\epsilon = (0, 0, 0)$$
$$w = 0,\ b = 1.1: \quad \tfrac{1}{2}\|w\|^2 = 0, \quad J = 0$$

Calculate and explain

Under this objective and tolerance, no more complicated slope is needed. A model passing exactly through all targets cannot improve the already-zero data loss or norm penalty.

Meaning

Zero training objective says the observed rows meet a specified tolerance. It does not prove all future wear predictions are accurate or that target variability is absent.

Compare absolute, squared, tube, and Huber penalties

Question and idea

Use the same residuals to compare preferences. Here Huber has threshold $\delta = 0.2$ and equals $r^2/2$ inside its quadratic region, then $\delta(|r| - \delta/2)$ outside.

$\lvert r\rvert$	Absolute	Squared	Tube $\epsilon = .2$	Huber $\delta = .2$
.1	.1	.01	0	.005
.5	.5	.25	.3	.08

Calculate and explain

Tube loss ignores a central interval; Huber retains a smooth quadratic response there. Squared loss gives large residuals relatively greater influence than linear-tail penalties.

Meaning

Numerically smaller losses across different columns do not imply a better predictor. The scales and units differ; evaluation must use one common task metric.

Two slacks represent the two sides of a regression tube

Idea

A linear SVR controls slope magnitude while paying for residuals outside either side. Two nonnegative slacks represent upward and downward excess separately.

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}\|w\|^2 + C\sum_i (\xi_i + \xi_i^*)$$

$$y_i - f(x_i) \le \epsilon + \xi_i, \quad f(x_i) - y_i \le \epsilon + \xi_i^*, \quad \xi_i, \xi_i^* \ge 0$$

Work through it

For $y = 2.5$, $f = 2.0$, and $\epsilon = 0.2$, the smallest slacks are $\xi = 0.3, \xi^* = 0$. If $y = 1.5$ instead, their roles reverse.

Interpretation

The classification label multiplier disappears. Regression targets are continuous and either residual direction may violate the tube.

Calculate both tube slacks for a batch of predictions

Question and idea

Use ξ for positive residual excess and ξ^* for negative residual excess. With epsilon 0.2, every row needs at most one strictly positive minimal slack.

$$y = (1, 2, 4), \quad f = (1.1, 2.6, 3.6), \quad r = (-.1, -.6, .4)$$
$$\xi = (0, 0, .2), \quad \xi^* = (0, .4, 0), \quad \sum_i (\xi_i + \xi_i^*) = .6$$

Calculate and explain

The first prediction needs no slack, the second exceeds the lower side by 0.4, and the third exceeds the upper side by 0.2. If the norm penalty is 2 and $C = 3$, the objective is 3.8.

Meaning

Two slack types represent directions, not two independent extra errors for each observation. Use the minimal feasible values when evaluating a fitted prediction.

Price outside-tube residuals when comparing two regression fits

Question and idea

Suppose two feasible regressors have norm penalties 0.5 and 2, and total excess errors 3 and 1. The preferred fit depends on the common excess-error price.

$$J_A = .5 + 3C, \qquad J_B = 2 + C$$

$$C = .25: \; J_A = 1.25 < J_B = 2.25; \qquad C = 2: \; J_B = 4 < J_A = 6.5$$

Calculate and explain

The crossover is $C = 0.75$. Below it the shorter direction with larger tolerated excess wins; above it the better-fitting but larger-norm candidate wins.

Meaning

Candidates must share targets, epsilon, feature scaling, and the same summation convention. Otherwise their total excesses cannot be compared as one objective.

Dual SVR predicts with signed coefficient differences

Idea

The two tube inequalities have multipliers α_i and α_i^*. Their difference supplies the signed contribution of a support row, without a binary class label.

$$f(x) = \sum_i (\alpha_i - \alpha_i^*) K(x_i, x) + b$$

$$0 \le \alpha_i, \alpha_i^* \le C, \qquad \sum_i (\alpha_i - \alpha_i^*) = 0$$

Work through it

A row strictly inside the tube has both multipliers zero under nondegenerate conditions. Rows on or outside an edge can contribute; outside rows commonly reach the relevant cap.

Interpretation

A positive or negative coefficient describes residual direction, not a positive or negative class. The intercept is recovered using the appropriate tube-edge condition.

Read the SVR dual objective with two coefficient families

Question and idea

Let $\beta_i = \alpha_i - \alpha_i^*$. The quadratic term penalizes function norm, the epsilon term charges both families, and observed target values reward signed coefficients.

$$\max \; W = -\tfrac{1}{2} \sum_{i,j} \beta_i \beta_j K_{ij} - \epsilon \sum_i (\alpha_i + \alpha_i^*) + \sum_i y_i \beta_i$$

$$0 \le \alpha_i, \alpha_i^* \le C, \qquad \sum_i \beta_i = 0$$

Calculate and explain

Unlike classification, the last term uses continuous targets y_i, not binary labels multiplying a single alpha. The two inequalities lead naturally to two possible residual directions.

Meaning

The coefficient difference predicts the function, but the epsilon penalty depends on the sum of nonnegative coefficients. Keeping only beta too early loses that structure.

Solve a two-point SVR in the dual, including epsilon

Question and idea

Use $(x, y) = (0, 0), (1, 1)$, a linear kernel, epsilon 0.1 and $C = 1$. Let the first row have $\alpha_0^* = a$ and the second $\alpha_1 = a$, giving $\beta = (-a, a)$.

$$w = a, \quad W(a) = -\tfrac{1}{2}a^2 - .2a + a = .8a - \tfrac{1}{2}a^2$$
$$W'(a) = .8 - a = 0 \Rightarrow a = .8, \quad w = .8, \quad b = .1$$

Calculate and explain

Predictions are 0.1 and 0.9, each exactly 0.1 away from its target. Data loss is zero; primal objective and dual optimum both equal 0.32.

Meaning

A regularized tube fit deliberately uses slope 0.8 rather than exact interpolation slope 1. Both coefficient directions are needed to balance the intercept condition.

Recover the SVR intercept from the correct tube edge

Question and idea

A free positive-residual support satisfies $y_i - f_i = \epsilon$; a free negative-residual support satisfies $f_i - y_i = \epsilon$. Each edge gives a different sign of epsilon in the intercept formula.

$$0 < \alpha_i < C: \quad b = y_i - \epsilon - \sum_j \beta_j K(x_j, x_i)$$

$$0 < \alpha_i^* < C: \quad b = y_i + \epsilon - \sum_j \beta_j K(x_j, x_i)$$

Calculate and explain

For the two-point solution $w = 0.8$, the row at 0 gives $b = 0 + 0.1 - 0 = 0.1$; the row at 1 gives $b = 1 - 0.1 - 0.8 = 0.1$.

Meaning

Using the classification intercept formula or one epsilon sign for both edges produces an inconsistent regression tube.

Calculate a kernel-SVR prediction from signed supports

Question and idea

Take support inputs 0 and 1, differences $\beta = (-0.5, 0.5)$, RBF gamma 1, and intercept 0.2. This explicitly supplied model illustrates prediction; it is not a claimed fitted optimum.

$$f(x) = -.5e^{-x^2} + .5e^{-(x-1)^2} + .2$$
$$f(.5) = .2, \qquad f(1) = -.5e^{-1} + .5 + .2 \approx .516$$

Calculate and explain

At the midpoint the equal similarities cancel. At 1 the second support contributes more, so the predicted continuous target increases. No binary-label sign is applied after the sum.

Meaning

Kernel-SVR output is a continuous quantity in target units. Its magnitude is not a probability, and its local support behavior must be validated.

C prices excess error; epsilon defines what counts as excess

Idea

For fixed predictions, enlarging epsilon reduces the number of nonzero loss terms. Increasing C makes the remaining excesses more expensive. Neither parameter is a general accuracy knob.

Residual $\lvert r \rvert$	Loss at $\epsilon = 0.2$	Loss at $\epsilon = 0.5$
0.1	0	0
0.4	0.2	0
0.8	0.6	0.3

Work through it

If targets are divided by 10, an epsilon of 0.2 in original units becomes 0.02 in scaled units. Restore predictions to original units before reporting MAE.

Interpretation

Larger tubes often yield fewer active rows, but the final fitted support count is data-dependent. Scaling the target changes parameter interpretation.

A coefficient cap can make the SVR intercept nonunique

Question and idea

Reuse the two-point linear SVR, epsilon 0.1, but set $C = 0.4$. The dual optimum is clipped at $a = 0.4$, so $w = 0.4$. A symmetric valid intercept is 0.3.

$$f(0) = .3, \quad f(1) = .7, \quad \sum \ell_\epsilon = .2 + .2 = .4$$

$$J = .5(.4)^2 + .4(.4) = .24 = W(.4)$$

Calculate and explain

Any intercept from 0.1 to 0.5 gives the same minimal total excess 0.4 for this slope. Both active coefficients are at their caps, so free-support edge equalities do not uniquely fix b.

Meaning

A solver may choose one point in a valid interval. Different intercepts here need not signal different optimized objective values.

Wider tubes can reduce slope before changing residual signs

Question and idea

For the two targets 0 and 1 at inputs 0 and 1, a sufficiently large C permits a zero-excess optimum. The shortest feasible slope depends on the tube overlap.

$$w = \max(0, 1 - 2\epsilon), \quad b = \epsilon \quad (0 \le \epsilon \le .5)$$
$$\epsilon = 0, .1, .4 : \quad (w, b) = (1, 0), (.8, .1), (.2, .4)$$

Calculate and explain

At epsilon 0.5, the constant prediction 0.5 fits both targets inside the tube with zero norm. Below 0.5, each solution touches opposing tube edges.

Meaning

This exact small example isolates the tolerance effect. General support counts and validation errors are data-dependent and need not follow one monotonic formula.

Scaling the target changes C as well as epsilon for equivalent fits

Question and idea

Divide targets and predictions by $s > 0$ while leaving input features unchanged. For an equivalent squared-norm-plus-summed-tube-loss objective, coefficients, intercept, tolerance, and price must all scale consistently.

$$w' = w/s, \quad b' = b/s, \quad \epsilon' = \epsilon/s, \quad C' = C/s$$
$$J/s^2 = \tfrac{1}{2}\|w'\|^2 + (C/s)\sum_i \ell_{\epsilon'}(y'_i - f'_i)$$

Calculate and explain

With $s = 5$, original epsilon 1 becomes 0.2 and original $C = 10$ becomes 2. Restoring the predictions by multiplication with 5 recovers the same original-unit solution.

Meaning

Changing only epsilon preserves tolerance units but not necessarily the full regularization tradeoff. This equivalence assumes the stated objective convention.

Forecast demand with a chronological validation design

Idea

For a bike-demand example, predict a future count using features available at the prediction time. Earlier periods train the model, later periods validate it, and the final period tests it.

$$\text{earlier training} \longrightarrow \text{later validation} \longrightarrow \text{future test}$$

Work through it

Tune C, gamma, and epsilon on time-respecting splits. Report MAE in demand units, compare a simple seasonal baseline, and inspect residuals across hour and weather conditions.

Interpretation

RBF predictions outside the training feature range are not guaranteed to extrapolate a trend. Future information in features invalidates the exercise.

Construct chronological folds with a future test block

Question and idea

For monthly demand, build validation folds that always train on earlier months. Reserve June as a final future test and do not use it to select any kernel or tolerance.

Fold	Training months	Validation months
1	January–March	April
2	January–April	May
Final	January–May	June: test only

Calculate and explain

Fit feature and target transforms inside each training block. Use weather information available at forecast time, not an observed daily summary obtained after the target period ends.

Meaning

Adjacent observations may share strong dependence. A chronological split tests the intended future-use question; random folds answer a different question.

Calculate MAE and RMSE without mixing their purposes

Question and idea

Evaluate the same original-unit residuals $r = (-1, 1, 0, 4)$. MAE averages absolute errors; RMSE squares before averaging and then returns to target units.

$$\mathrm{MAE} = \frac{1 + 1 + 0 + 4}{4} = 1.5$$

$$\mathrm{RMSE} = \sqrt{\frac{1 + 1 + 0 + 16}{4}} = \sqrt{4.5} \approx 2.121$$

Calculate and explain

The single residual 4 affects RMSE more strongly. Both are reported in the target's units after the square root, while raw MSE has squared units.

Meaning

A chosen training epsilon does not make MAE zero inside the tube. Evaluation can still count all actual prediction errors, including tolerated training residuals.

Compare SVR with a seasonal baseline on paired targets

Question and idea

This constructed example uses targets $y = (10, 10, 20)$, baseline predictions $(10, 12, 18)$, and SVR predictions $(11, 10, 18)$. Compare both on the same three rows.

$$\mathrm{MAE}_{\mathrm{base}} = (0 + 2 + 2)/3 = 4/3$$
$$\mathrm{MAE}_{\mathrm{SVR}} = (1 + 0 + 2)/3 = 1$$

Calculate and explain

SVR improves the average by 1/3 here, but it is worse on the first row and ties on the third. Paired error differences expose what the average hides.

Meaning

Three teaching rows are not evidence of superiority in real demand forecasting. A meaningful comparison needs representative future blocks and an uncertainty-aware interpretation.

Respect a nonnegative target without pretending SVR is a count model

Question and idea

An unconstrained SVR can predict below zero. If the task's target is truly nonnegative, a predeclared output rule $g(f) = \max(0, f)$ can enforce that range.

$$y = 0,\ f = -.5: \quad |y - f| = .5, \quad |y - g(f)| = 0$$
$$y = 2,\ f = 1.5: \quad g(f) = 1.5$$

Calculate and explain

Clipping affects only negative predictions in this rule. Evaluate the entire model-plus-output rule consistently on validation and test data, and disclose it in the prediction definition.

Meaning

Range enforcement does not give integer counts or a probabilistic count distribution. It cannot repair missing future information or a shifted demand process.

Choose the model from the question, not from its name

Target first

One instrument may classify a defect or predict wear. The measurements can match while the outputs, losses, and error units differ.

	Defect classification	Wear regression
Target	$y \in \{-1, +1\}$	Continuous wear units
Data penalty	$\max(0, 1 - yf)$	$\max(0, \lvert y - f \rvert - \epsilon)$
Design	Class cost, C, kernel	Tolerance, C, kernel
Evaluation	Recall, balanced accuracy	MAE, residual patterns

Reasoned choice

Begin with a scaled linear baseline; add nonlinear similarity only when comparable validation supports it. Shared optimization structure does not erase label meaning, output units, or the cost of an error.

One instrument needs two distinct decision explanations

Question and idea

An item has a defect label and a measured wear value. A classifier answers a category question; an SVR answers a continuous tolerance question. Their errors must be evaluated separately.

$$y_{\text{class}} = +1, \quad f_{\text{class}} = .4 : \quad \ell_{\text{hinge}} = .6$$
$$y_{\text{wear}} = 2.5, \quad f_{\text{wear}} = 2.0, \quad \epsilon = .2 : \quad \ell_{\epsilon} = .3$$

Calculate and explain

The first prediction has the correct class sign but insufficient margin. The second has 0.5 mm actual error and 0.3 mm penalized excess. Neither score can substitute for the other output.

Meaning

A defect-recall policy and an acceptable wear error are different requirements. Shared input sensors do not merge the target definitions or the units of loss.

Explain the complete chain and carry similarity forward

Idea

The lecture connects geometric distance, normalized constraints, penalized violations, row coefficients, kernel similarity, and a regression tolerance tube. Each new term solves a specific earlier limitation.

1. For $w = (3, 4), b = -5, x = (1, 1)$, find score and distance.
2. Interpret $yf = 0.4$: class sign, slack, and hinge loss.
3. Explain why $0 < \alpha_i < C$ identifies a free support.
4. Compute RBF similarity at $d = 2, \gamma = 0.5$.
5. For $|r| = 0.5, \epsilon = 0.2$, compute SVR loss.

Work through it

Answers: 2 and 0.4; correct sign with 0.6 violation; KKT gives $yf = 1$; $e^{-2} \approx 0.135$; 0.3.

Interpretation

Next lecture keeps representations and similarities but removes target labels: useful clusters must be defined and evaluated differently.

Solve a final tube case and identify the validation limit

Question and idea

A fitted linear regressor is $f(x) = 0.8x + 0.1$ with epsilon 0.1. Compute predictions and losses for $(x, y) = (0, 0), (1, 1), (2, 2.4)$; its training inputs originally covered only 0 to 1.

$$f = (.1, .9, 1.7), \quad |r| = (.1, .1, .7)$$
$$\ell_\epsilon = (0, 0, .6), \qquad \mathrm{MAE} = .9/3 = .3$$

Calculate and explain

The first two observations lie on the tube edges. The third is outside and also outside the original input range; its 0.6 excess should not be hidden by the zero losses of the training-like points.

Meaning

The arithmetic is valid for this explicitly linear model. Whether extrapolation works operationally needs future-range evidence; a kernel model can behave differently outside its supports.